

STRANGE DUALITY FOR ABELIAN SURFACES

RYAN KANNANAIKAL

ABSTRACT. This is where I put the abstract.

CONTENTS

1. Intro	1
References	2

1. INTRO

[1] [10] [3] [4] [5] [6] [7] [8] [2] [9] [11] [12] [13] [15] [14] [16] [17] [18] [19] [20] [21] [22] [24] [23] [25]
[28] [26] [27] [29] [30] [31] [32] [33] [34] [35] [37] [36] [38] [40] [39] [41] [42] [43] [44] [45] [46] [47]

REFERENCES

- [1] M. F. Atiyah, *Vector bundles over an elliptic curve*, Proc. London Math. Soc. (3) **7** (1957), 414–452, DOI [10.1112/plms/s3-7.1.414](#). MR 0131423
- [2] A. Bayer and E. Macrì, *MMP for moduli of sheaves on K3s via wall-crossing: nef and movable cones, Lagrangian fibrations*, Invent. Math. **198** (2014), no. 3, 505–590, DOI [10.1007/s00222-014-0501-8](#). MR 3279532
- [3] A. Beauville, *Variétés kähleriennes dont la première classe de Chern est nulle*, J. Differential Geom. **18** (1983), no. 4, 755–782 (1984). MR 0730926
- [4] A. Beauville, *Vector bundles on curves and generalized theta functions: recent results and open problems*, *Current topics in complex algebraic geometry (Berkeley, CA, 1992/93)*, Math. Sci. Res. Inst. Publ. vol. 28, Cambridge Univ. Press, Cambridge, 1995, pp. 17–33. MR 1397056
- [5] P. Belkale, *The strange duality conjecture for generic curves*, J. Amer. Math. Soc. **21** (2008), no. 1, 235–258, DOI [10.1090/S0894-0347-07-00569-3](#). MR 2350055
- [6] P. Belkale, *Strange duality and the Hitchin/WZW connection*, J. Differential Geom. **82** (2009), no. 2, 445–465. MR 2520799
- [7] M. Bernardara and G. Hein, *The Euclid-Fourier-Mukai algorithm for elliptic surfaces*, Asian J. Math. **18** (2014), no. 2, 345–364, DOI [10.4310/AJM.2014.v18.n2.a8](#). MR 3217640
- [8] C. Birkenhake and H. Lange, *Complex abelian varieties*, 2nd ed., Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], vol. 302, Springer-Verlag, Berlin, 2004, DOI [10.1007/978-3-662-06307-1](#). MR 2062673
- [9] B. Bolognese, A. Marian, D. Oprea, and K. Yoshioka, *On the strange duality conjecture for abelian surfaces II*, J. Algebraic Geom. **26** (2017), no. 3, 475–511, DOI [10.1090/jag/685](#). MR 3647791
- [10] T. Bridgeland, *Fourier-Mukai transforms for elliptic surfaces*, J. Reine Angew. Math. **498** (1998), 115–133, DOI [10.1515/crll.1998.046](#). MR 1629929
- [11] E. Colombo and G. P. Pirola, *Some density results for curves with nonsimple Jacobians*, Math. Ann. **288** (1990), no. 1, 161–178, DOI [10.1007/BF01444527](#). MR 1070930
- [12] G. Danila, *Sections du fibré déterminant sur l’espace de modules des faisceaux semi-stables de rang 2 sur le plan projectif*, Ann. Inst. Fourier (Grenoble) **50** (2000), no. 5, 1323–1374, DOI [10.5802/aif.1795](#). MR 1800122
- [13] G. Danila, *Résultats sur la conjecture de dualité étrange sur le plan projectif*, Bull. Soc. Math. France **130** (2002), no. 1, 1–33, DOI [10.24033/bsmf.2410](#). MR 1906190
- [14] R. Donagi and L. W. Tu, *Theta functions for $SL(n)$ versus $GL(n)$* , Math. Res. Lett. **1** (1994), no. 3, 345–357, DOI [10.4310/MRL.1994.v1.n3.a6](#). MR 1302649
- [15] J. M. Drezet and M. S. Narasimhan, *Groupe de Picard des variétés de modules de fibrés semi-stables sur les courbes algébriques*, Invent. Math. **97** (1989), no. 1, 53–94, DOI [10.1007/BF01850655](#). MR 0999313
- [16] G. Ellingsrud, L. Göttsche, and M. Lehn, *On the cobordism class of the Hilbert scheme of a surface*, J. Algebraic Geom. **10** (2001), no. 1, 81–100. MR 1795551
- [17] R. Friedman, *Rank two vector bundles over regular elliptic surfaces*, Invent. Math. **96** (1989), no. 2, 283–332, DOI [10.1007/BF01393965](#). MR 0989699
- [18] R. Friedman, *Algebraic surfaces and holomorphic vector bundles*, Universitext, Springer-Verlag, New York, 1998, DOI [10.1007/978-1-4612-1688-9](#). MR 1600388
- [19] W. Fulton, *Intersection theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) [Results in Mathematics and Related Areas (3)], vol. 2, Springer-Verlag, Berlin, 1984, DOI [10.1007/978-3-662-02421-8](#). MR 0732620

- [20] G. van der Geer, *The cohomology of the moduli space of abelian varieties*, *Handbook of moduli. Vol. I*, Adv. Lect. Math. (ALM), vol. 24, Int. Press, Somerville, MA, 2013, pp. 415–457. MR 3184169
- [21] N. J. Hitchin, *Flat connections and geometric quantization*, *Comm. Math. Phys.* **131** (1990), no. 2, 347–380. MR 1065677
- [22] K. Hulek, C. Kahn, and S. H. Weintraub, *Moduli spaces of abelian surfaces: compactification, degenerations, and theta functions*, *De Gruyter Expositions in Mathematics*, vol. 12, Walter de Gruyter & Co., Berlin, 1993, DOI [10.1515/9783110891928](https://doi.org/10.1515/9783110891928). MR 1257185
- [23] D. Huybrechts, *Fourier-Mukai transforms in algebraic geometry*, *Oxford Mathematical Monographs*, The Clarendon Press, Oxford University Press, Oxford, 2006, DOI [10.1093/acprof:oso/9780199296866.001.0001](https://doi.org/10.1093/acprof:oso/9780199296866.001.0001). MR 2244106
- [24] D. Huybrechts and M. Lehn, *The geometry of moduli spaces of sheaves*, 2nd ed., *Cambridge Mathematical Library*, Cambridge University Press, Cambridge, 2010, DOI [10.1017/CB09780511711985](https://doi.org/10.1017/CB09780511711985). MR 2665168
- [25] E. Izadi, *Density and completeness of subvarieties of moduli spaces of curves or abelian varieties*, *Math. Ann.* **310** (1998), no. 2, 221–233, DOI [10.1007/s002080050146](https://doi.org/10.1007/s002080050146). MR 1602079
- [26] J. Le Potier, *Fibré déterminant et courbes de saut sur les surfaces algébriques*, *Complex projective geometry (Trieste, 1989/Bergen, 1989)*, *London Math. Soc. Lecture Note Ser.* vol. 179, Cambridge Univ. Press, Cambridge, 1992, pp. 213–240, DOI [10.1017/CB09780511662652.016](https://doi.org/10.1017/CB09780511662652.016). MR 1201385
- [27] J. Le Potier, *Dualité étrange sur le plan projectif*, *Lectures at Luminy*, 1996.
- [28] J. Li, *Algebraic geometric interpretation of Donaldson’s polynomial invariants*, *J. Differential Geom.* **37** (1993), no. 2, 417–466. MR 1205451
- [29] S. Makarova, *Moduli spaces of stable sheaves over quasi-polarized surfaces, and the relative strange duality morphism*, *Épjournal Géom. Algébrique* **5** (2021), Art. 19, 15, DOI [10.46298/epiga.2021.7174](https://doi.org/10.46298/epiga.2021.7174). MR 4354034
- [30] A. Marian and D. Oprea, *The level-rank duality for non-abelian theta functions*, *Invent. Math.* **168** (2007), no. 2, 225–247, DOI [10.1007/s00222-006-0032-z](https://doi.org/10.1007/s00222-006-0032-z). MR 2289865
- [31] A. Marian and D. Oprea, *A tour of theta dualities on moduli spaces of sheaves, Curves and abelian varieties*, *Contemp. Math.* vol. 465, Amer. Math. Soc., Providence, RI, 2008, pp. 175–201, DOI [10.1090/conm/465/09103](https://doi.org/10.1090/conm/465/09103). MR 2457738
- [32] A. Marian and D. Oprea, *Sheaves on abelian surfaces and strange duality*, *Math. Ann.* **343** (2009), no. 1, 1–33, DOI [10.1007/s00208-008-0262-z](https://doi.org/10.1007/s00208-008-0262-z). MR 2448439
- [33] A. Marian and D. Oprea, *Generic strange duality for K3 surfaces*, *Duke Math. J.* **162** (2013), no. 8, With an appendix by Kota Yoshioka, 1463–1501, DOI [10.1215/00127094-2208643](https://doi.org/10.1215/00127094-2208643). MR 3079253
- [34] A. Marian and D. Oprea, *On the strange duality conjecture for abelian surfaces*, *J. Eur. Math. Soc. (JEMS)* **16** (2014), no. 6, 1221–1252, DOI [10.4171/JEMS/459](https://doi.org/10.4171/JEMS/459). MR 3226741
- [35] A. Marian and D. Oprea, *On Verlinde sheaves and strange duality over elliptic Noether-Lefschetz divisors*, *Ann. Inst. Fourier (Grenoble)* **64** (2014), no. 5, 2067–2086, DOI [10.5802/aif.2904](https://doi.org/10.5802/aif.2904). MR 3330931
- [36] S. Mukai, *Semi-homogeneous vector bundles on an Abelian variety*, *J. Math. Kyoto Univ.* **18** (1978), no. 2, 239–272, DOI [10.1215/kjm/1250522574](https://doi.org/10.1215/kjm/1250522574). MR 0498572
- [37] S. Mukai, *Duality between $D(X)$ and $D(\hat{X})$ with its application to Picard sheaves*, *Nagoya Math. J.* **81** (1981), 153–175. MR 0607081
- [38] S. Mukai, *Symplectic structure of the moduli space of sheaves on an abelian or K3 surface*, *Invent. Math.* **77** (1984), no. 1, 101–116, DOI [10.1007/BF01389137](https://doi.org/10.1007/BF01389137). MR 0751133

- [39] D. Mumford, *On the equations defining abelian varieties. I*, Invent. Math. **1** (1966), 287–354, DOI [10.1007/BF01389737](#). MR 0204427
- [40] D. Mumford, *Abelian varieties*, Tata Institute of Fundamental Research Studies in Mathematics, vol. 5, Tata Institute of Fundamental Research, Bombay; by Oxford University Press, London, 1970. MR 0282985
- [41] K. G. O’Grady, *The weight-two Hodge structure of moduli spaces of sheaves on a K3 surface*, J. Algebraic Geom. **6** (1997), no. 4, 599–644. MR 1487228
- [42] C. Okonek, M. Schneider, and H. Spindler, *Vector bundles on complex projective spaces*, Progress in Mathematics, vol. 3, Birkhäuser, Boston, MA, 1980. MR 0561910
- [43] S. Ramanan, *The moduli spaces of vector bundles over an algebraic curve*, Math. Ann. **200** (1973), 69–84, DOI [10.1007/BF01578292](#). MR 0325615
- [44] B. Russo and M. Teixidor i Bigas, *On a conjecture of Lange*, J. Algebraic Geom. **8** (1999), no. 3, 483–496. MR 1689352
- [45] L. Scala, *Dualité étrange de Le Potier et cohomologie du schéma de Hilbert ponctuel d’une surface*, Gaz. Math. (2007), no. 112, 53–65. MR 2319915
- [46] S. Yanagida and K. Yoshioka, *Semi-homogeneous sheaves, Fourier-Mukai transforms and moduli of stable sheaves on abelian surfaces*, J. Reine Angew. Math. **684** (2013), 31–86, DOI [10.1515/crelle-2011-0010](#). MR 3181556
- [47] K. Yoshioka, *Moduli spaces of stable sheaves on abelian surfaces*, Math. Ann. **321** (2001), no. 4, 817–884, DOI [10.1007/s002080100255](#). MR 1872531

DEPARTMENT OF MATHEMATICS, NORTHEASTERN UNIVERSITY, 43 LEON STREET, BOSTON, MA 02115, USA
 Email address: kannanaikal.r@northeastern.edu